

LAB EXPERIMENT 4

Implement and analyze the following Graph Algorithm problems and demonstrate their working. Take input data from the user.

PART 1

Write a program to find the Minimum Spanning Tree (MST) of a weighted graph using Prim's and Kruskal's algorithm.

- Display the edges selected in the MST along with their weights, for both algorithms.
- Display the total minimum cost of the spanning tree.
- Compare the edges chosen and the order in which each algorithm selects them.

Practice on LeetCode (free):

- LeetCode: 1584. Min Cost to Connect All Points — core MST problem, solvable with Prim's or Kruskal's algorithm.
- LeetCode: 1319. Number of Operations to Make Network Connected — Union-Find practice closely related to Kruskal's algorithm.
- LeetCode: 1489. Find Critical and Pseudo-Critical Edges in Minimum Spanning Tree — advanced MST analysis built on Kruskal's algorithm.

PART 2

Write a program to find the Single Source Shortest Path in a weighted graph using Dijkstra's algorithm.

- Display the shortest distance from the source vertex to every other vertex.
- Display the path taken to reach each vertex from the source.
- Ensure the program works correctly on graphs with non-negative edge weights.

Practice on LeetCode (free):

- LeetCode: 743. Network Delay Time — classic Dijkstra's algorithm application.
- LeetCode: 1631. Path With Minimum Effort — Dijkstra's algorithm with a priority queue over path cost.
- LeetCode: 1976. Number of Ways to Arrive at Destination — Dijkstra's algorithm combined with shortest-path counting.